

# Dany Binu Luke

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🔗 Codeforces    🌐 Portfolio

## Education

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**Bachelor of Technology, Computer Science and Engineering** *Aug 2023 – Present*  
*National Institute of Technology, Surat*

**High School Diploma** *Apr 2009 – Mar 2023*  
*United Indian School, Kuwait*

## Projects

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**MicroState - Real-Time Market Microstructure Analytics System** *Python, FastAPI, React, Docker*

- Engineered a real-time system reconstructing Level-2 limit order books with event-driven microstructure analysis achieving 1-second latency on 50+ cryptocurrency data streams; implemented multi-timescale ML architecture with 92% prediction accuracy.
- Created full-stack dashboard with React and FastAPI supporting 100+ concurrent users via WebSocket streaming with real-time visualization of 15+ market metrics; containerized with Docker handling 1000+ requests/second.

**SentiStock AI - Stock Sentiment Analysis and Prediction** *Python, NLP, Machine Learning*

- Constructed machine learning pipeline processing 500+ financial news articles daily using transformer-based sentiment analysis achieving 87% classification accuracy for automated market trend prediction.
- Implemented real-time data ingestion forecasting price movements with 78% directional accuracy, delivering actionable insights faster than manual analysis approaches.

**Vaani - Speech-to-Text Summarizer** *Torch, Transformers, Librosa, Gradio*

- Architected AI-powered platform integrating Automatic Speech Recognition, real-time translation, and abstractive summarization for Indian languages with 94% transcription accuracy on Hindi audio.
- Deployed interactive web interface currently processing 100+ audio files monthly; expanding multilingual support to Malayalam which is in development.

## Skills

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**Programming Languages:** Python, C, C++, Java, JavaScript, MySQL

**CS Fundamentals:** Object-Oriented Programming, Database Management System, Data Structures, Computer Networks, Operating Systems

**Tools/Frameworks:** HTML5, CSS, PyTorch, OpenCV, YOLO, HuggingFace Transformers, NumPy, Pandas, NLP, MERN (MongoDB, ExpressJS, ReactJS, NodeJS), NextJS, Framer Motion, Docker, KuberNetes, REST APIs

**Additional Skills:** Teamwork and Project Management, Effective written and verbal communication.

## Achievements

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**Google Cloud Study Jams - Gold League:** Completed comprehensive training covering Google Cloud services including Compute, Storage, API Gateway, Looker, Dataplex, and Pub/Sub; developed 5+ GenAI applications using Vertex AI and Gemini API with Streamlit.

**NPTEL - Privacy and Security in Online Social Media:** Achieved top 5% ranking under Prof. Ponnu-rangam Kumaraguru; demonstrated expertise in online privacy risks, social engineering attacks, and ethical data handling practices.